Antibiotic Resistance Gene Co-occurrence Network Topology Under Climate Gradients: A Gene-Level Network Analysis

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Abstract

Antibiotic resistance genes (ARGs) in environmental metagenomes do not accumulate in isolation; they are co-selected, co-transferred, and co-regulated in ways that produce structured co-occurrence patterns. Whether these patterns vary systematically with climate zone remains largely unexplored. This study constructs weighted ARG co-occurrence networks for three major climate zones—tropical, arid, and temperate—using the mARG dataset (Martiny *et al.*) comprising 119,232 metagenomic samples, with climate zone assigned via the Beck *et al.* (2023) Köppen-Geiger raster at 0.1° resolution. Pairwise Spearman correlations between 156 ARGs passing a 5% presence filter were used to define edges ($r \geq 0.3$, $p < 0.05$). Louvain community detection revealed modularity scores of $Q = 0.41$ (tropical), $Q = 0.26$ (arid), and $Q = 0.31$ (temperate), indicating zone-specific variation in network compartmentalisation. Hub ARGs identified by betweenness centrality differed across zones, with `tet(M)` recurring in both tropical and arid networks. A Kruskal-Wallis test on betweenness distributions across zones yielded $H = 3.22$, $p = 0.20$ (not significant), suggesting that while the *identity* of hub ARGs differs by zone, the overall structural centrality distribution is conserved. These findings provide a network-level perspective on how climate context may shape the ecology of antibiotic resistance.

Keywords: Antibiotic Resistance Genes, Co-occurrence Network, Köppen-Geiger Climate Classification, Community Detection, Louvain Algorithm, Betweenness Centrality, Network Topology, Environmental Metagenomics

1 Introduction

The global rise of antibiotic resistance constitutes one of the most urgent public-health challenges of the twenty-first century. Antibiotic resistance genes (ARGs) are the molecular currency of this crisis: they encode biochemical mechanisms—enzymatic inactivation, efflux pumps, target modification—that render antibiotics ineffective [?]. Environmental metagenomics has demonstrated that ARGs are pervasive across soils, sediments, and waterways, raising concern about the "resistome" as a reservoir from which clinically relevant resistance elements may emerge [?].

ARGs are rarely found in isolation. They are frequently co-localised on mobile genetic elements such as plasmids and integrons, creating co-selection pressure that drives their co-occurrence in metagenomic samples [?]. Network science offers a natural framework for studying these co-occurrence patterns. By treating ARGs as nodes and statistically significant co-occurrences as edges, one can characterise the global structure of resistance gene ecology—identifying hub genes that link distinct resistance clusters, and detecting communities of genes that tend to co-occur together.

Climate zone is a plausible predictor of ARG community structure. Temperature and precipitation shape microbial community composition, horizontal gene transfer rates, and the selective pressure from antibiotic use in agriculture [? ?]. Yet to our knowledge, no study has directly compared ARG co-occurrence *network topology* across climate zones using a large, geographically diverse metagenomic dataset.

This study addresses that gap. Using the mARG dataset—the largest unified ARG abundance table compiled from published environmental metagenomic studies—we assign climate zone labels via the high-resolution Köppen-Geiger raster of Beck *et al.* (2023), construct weighted co-occurrence networks for tropical, arid, and temperate zones, and compare their topology through community detection and centrality analysis.

2 Review of Related Literature

2.1 The Global Burden of Antimicrobial Resistance

Antimicrobial resistance has emerged as one of the defining public health crises of the twenty-first century. The World Health Organization classifies AMR among the top ten threats to global health, and surveillance data indicate that resistant infections now cause millions of deaths annually [?]. Drug-resistant pathogens complicate the treatment of common bacterial infections, surgical procedures, and cancer chemotherapy, eroding the clinical foundations of modern medicine. The burden is not distributed evenly: low- and middle-income countries, which already face constrained healthcare infrastructure, carry a disproportionate share of resistance-attributable mortality [?].

The environmental reservoir plays a central, if underappreciated, role in this crisis. Antibiotic resistance genes are ancient and pervasive features of microbial genomes, predating the clinical antibiotic era by millions of years [?]. Soils, sediments, and water bodies harbour vast and diverse resistomes whose composition reflects both natural selective pressures and the accumulated legacy of antibiotic use in medicine, agriculture, and aquaculture. Environmental ARGs constitute a standing reservoir from which clinically relevant resistance elements can be recruited into pathogens through horizontal gene transfer [?].

2.2 Climate Change as a Driver of Antimicrobial Resistance

Growing evidence links climatic variables to the abundance, diversity, and spread of environmental ARGs. A systematic scoping review of the climate-AMR literature found consistent associations between rising temperature and increased resistance rates across multiple pathogen species and geographic settings [?]. The biological mechanisms proposed to explain these associations include accelerated bacterial growth at higher temperatures, increased rates of horizontal gene transfer through bacterial conjugation, and heat-induced stress responses that upregulate efflux pumps and other resistance mechanisms [?].

Extreme weather events compound these pressures in a different way. Floods and heavy rainfall damage sanitation infrastructure and contaminate water sources, releasing ARGs from agricultural and wastewater reservoirs into the broader environment [?]. Droughts concentrate resistant bacteria and their genetic cargo in shrinking water bodies, increasing the probability of gene transfer events. These pathways position climate change not merely as a background driver of AMR but as an active amplifier of the spread and persistence of resistance genes across environments and hosts [?].

Despite this growing body of evidence, the relationship between climate and the structural organisation of ARG communities at the gene-network level remains largely uncharacterised. Most existing studies examine ARG abundance or resistance rates as scalar quantities, without capturing the co-occurrence relationships that reflect the ecological and genetic linkages among resistance determinants.

2.3 ARG Co-occurrence Networks in Environmental Metagenomics

Network science has emerged as a productive framework for studying the co-occurrence patterns of ARGs in environmental metagenomes. By representing ARGs as nodes and statistically significant co-occurrence events as edges, researchers can model the collective ecology of the environmental resistome, rather than examining individual genes in isolation. Early applications of this approach in diverse environments, including soil, freshwater, and marine sediments, established that ARG co-occurrence networks share structural properties with other biological networks: sparse but non-trivial connectivity, modular organisation into gene communities, and the presence of highly connected hub genes that link otherwise distinct clusters [**?**].

The co-occurrence of ARGs in the same environmental sample most commonly reflects physical co-localisation on shared mobile genetic elements such as plasmids, integrons, and transposons [**?**]. When two genes are repeatedly found together across geographically diverse samples, it suggests that they are co-selected and co-transferred as a functional unit. This makes co-occurrence network topology an indirect but tractable proxy for the genetic architecture of the resistome. Subsequent studies applying network analysis to global metagenomic datasets confirmed that co-occurrence patterns are highly context-dependent: ARG communities in sewage, agricultural soil, and pristine environments show distinct structural profiles, consistent with the influence of antibiotic selection pressure and microbial community composition on network organisation [**?**].

The use of large, unified ARG abundance tables, such as the mARG dataset compiled by Martiny *et al.*, makes it possible to extend this approach to the climate gradient question. Analysing co-occurrence structure across samples grouped by climate zone allows the structural properties of the resistome network to be compared under systematically different environmental conditions, complementing and extending the scalar abundance analyses that dominate the current climate-AMR literature.

2.4 Gap in the Literature

Despite parallel advances in climate-AMR research and resistome network analysis, no published study has directly compared ARG co-occurrence network topology across major climate zones using a large, geographically diverse, and uniformly processed metagenomic dataset. Existing network studies are typically restricted to single environment types or single geographic regions. Existing climate-AMR studies focus on resistance rates rather than network structure. The present study addresses this intersection, asking not simply whether climate zone correlates with ARG abundance, but whether it reshapes the topological organisation of the resistance gene co-occurrence network.

3 Methods

3.1 Dataset

We used the mARG dataset published by Martiny *et al.* (Zenodo accession 6919377), which aggregates ARG abundance tables from 119,232 environmental metagenomic samples. The dataset provides, for each sample-ARG pair, the fragment count aligned to a reference ARG sequence (`fragmentCountAln`), the reference sequence length, and the total number of trimmed reads (`trimmed_fragments`). Geographic coordinates were parsed from the accompanying `metadata.tsv` file, which records sample location in degree-minute format (e.g. "18.05 S 47.03 E").

Abundance was normalised to Reads Per Kilobase per Million mapped reads (RPKM):

$$\mathrm{RPKM}_{ij} = \frac{\mathrm{fragmentCountAln}_{ij} \times 10^6}{\left(\mathrm{refLength}_j/1000\right) \times \mathrm{trimmedFragments}_i} \tag{1}$$

where i indexes samples and j indexes ARG sequences. This normalisation accounts for both gene length bias and variation in sequencing depth. The resulting wide-format matrix has dimensions 119,232 samples $\times$ 2,999 ARGs.

3.2 Climate Zone Assignment

Each sample was assigned to a major climate class using the Beck *et al.* (2023) Köppen-Geiger raster at 0.1° spatial resolution (Figshare dataset 21789074). The raster encodes 30 climate types as integer codes; we mapped these to three analytical zones:

- **Tropical** (codes 1–3, class A)
- **Arid** (codes 4–7, class B)
- **Temperate** (codes 8–28, classes C and D merged)
- Polar (codes 29–30, class E) — excluded

Point queries were performed in Python using `rasterio`. Samples with coordinates outside the raster extent or returning a no-data value were discarded. This yielded 52,064 samples across the three zones (Table **??**).

3.3 ARG Filtering

ARGs present in fewer than 5% of samples (i.e. fewer than $0.05 \times 52{,}064 \approx 2{,}603$ samples) were removed as insufficiently prevalent to construct meaningful co-occurrence edges. This reduced the feature set from 2,999 to **156 ARGs**.

Table 1: Sample counts per climate zone after Köppen-Geiger assignment.

Climate Zone	Samples	Fraction (%)
Temperate	44,601	85.7
Tropical	4,443	8.5
Arid	3,020	5.8
Total	**52,064**	**100.0**

3.4 Co-occurrence Network Construction

For each climate zone, we computed all pairwise Spearman rank correlations between the 156 retained ARG abundance vectors. An undirected, weighted edge was added between ARG i and ARG j if and only if:

$$r_s(i,j) \geq 0.3 \quad \text{and} \quad p < 0.05 \tag{2}$$

where r_s is the Spearman correlation coefficient and p is its two-tailed p-value. The threshold $r \geq 0.3$ follows established practice in microbiome co-occurrence network studies [?] and excludes weakly correlated pairs that may reflect chance associations. Only positive correlations were retained, as negative correlations are ecologically ambiguous in this context (competitive exclusion *vs.* differential habitat preference). Networks were implemented as weighted undirected graphs in Python using `NetworkX` and exported as `.gexf` files.

3.5 Community Detection

Community structure was assessed using two algorithms:

1. **Louvain algorithm** (primary) [?]: a greedy modularity-maximisation heuristic that iteratively reassigns nodes to communities to improve the modularity score Q, defined as
$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \tag{3}$$
where A_{ij} is the edge weight, k_i is the weighted degree of node i, m is the total edge weight, c_i is the community assignment of node i, and δ is the Kronecker delta. Values of $Q > 0.3$ are conventionally regarded as indicating meaningful community structure [?].

2. **Girvan-Newman algorithm** (comparison) [?]: a divisive hierarchical approach that removes edges with the highest betweenness centrality iteratively; the first-level partition was used.

3.6 Centrality Metrics

Three node-level centrality measures were computed per zone:

1. **Degree centrality**: fraction of other nodes to which a given ARG is connected.
2. **Betweenness centrality**: proportion of shortest paths in the network that pass through a given ARG, normalised by the total number of pairs. Hub ARGs with high betweenness act as bridges between co-occurrence communities.
3. **Clustering coefficient**: probability that two neighbours of a given ARG are themselves connected, reflecting local clustering.

3.7 Cross-zone Comparison

To test whether betweenness centrality distributions differ significantly across climate zones, we applied the Kruskal-Wallis H-test [?], a non-parametric one-way analysis of variance appropriate for non-normal distributions. Significance was set at $\alpha = 0.05$.

4 Results

4.1 Network Topology

All three climate zone networks contained 155 nodes (one ARG with no significant correlations was isolated in each zone and contributed no edges). Network statistics are summarised in Table ??.

Table 2: Summary topology of ARG co-occurrence networks per climate zone.

Zone	Edges	Density	Avg. Clustering	Components
Tropical	1,730	0.145	0.315	24
Arid	2,292	0.192	0.326	15
Temperate	1,960	0.164	0.323	18

The arid zone network was the most densely connected (density = 0.192) with the fewest connected components (15), suggesting a more integrated resistance gene ecology. The tropical network was the sparsest (density = 0.145) with the most components (24), consistent with a more partitioned co-occurrence structure. Network visualisations are shown in Figure ??.

4.2 Community Detection

Louvain community detection results are presented in Table ??. The tropical zone exhibited the highest modularity ($Q = 0.41$), indicating strongly demarcated ARG communinetwork_tropical.png

Figure 1: (a) **Tropical Climate Zone** — 1,730 edges, $Q = 0.41$, Louvain communities: 27. Nodes are coloured by community and sized by betweenness centrality; labels shown for the top-12 hub ARGs by betweenness. Edge width is proportional to Spearman r.

network_arid.png

Figure 2: (b) **Arid Climate Zone** — 2,292 edges, $Q = 0.26$, Louvain communities: 20.

network_temperate.png

Figure 3: (c) **Temperate Climate Zone** — 1,960 edges, $Q = 0.31$, Louvain communities: 24.

ties. The arid zone fell below the conventional threshold of 0.3 ($Q = 0.26$), suggesting a more integrated, less compartmentalised co-occurrence network. The temperate zone was intermediate ($Q = 0.31$). Girvan-Newman partitioning produced qualitatively similar community counts.

Table 3: Community detection results per climate zone.

Zone	Louvain Communities	Modularity Q	GN Communities
Tropical	27	**0.4135**	25
Arid	20	0.2601	16
Temperate	24	0.3083	19

4.3 Hub ARGs by Betweenness Centrality

Table ?? lists the top five ARGs by betweenness centrality in each zone. `tet(M)` variants appear as primary hubs in both the tropical and arid zones, suggesting that this mobile tetracycline-resistance gene occupies a structurally central bridging role across ecologically distinct environments. In contrast, the temperate zone is dominated by `lnu(B)` and `aph(6)-Id`, genes conferring lincosamide and aminoglycoside resistance respectively, which were not among the top hubs in tropical or arid networks. `lsa(E)`, conferring streptogramin and lincosamide resistance, emerged as a top hub in both arid and temperate zones.

Table 4: Top 5 hub ARGs by betweenness centrality per climate zone.

Zone	ARG	Betweenness	Degree Centr.
	`tet(M)_13_AM990992`	0.060	0.299
	`tet(Q)_4_Z21523`	0.045	0.325
Tropical	`tet(M)_12_FR671418`	0.044	0.247
	`erm(B)_10_U86375`	0.039	0.143
	`tet(O)_3_Y07780`	0.030	0.344
	`aadA2_1_NC_010870`	0.081	0.182
	`lsa(E)_1_JX560992`	0.063	0.240
Arid	`tet(M)_13_AM990992`	0.061	0.331
	`lnu(A)_1_M14039`	0.056	0.032
	`aph(3'')-Ib_5_AF321551`	0.040	0.182
	`lnu(B)_2_JQ861959`	0.094	0.273
	`aph(6)-Id_1_M28829`	0.079	0.149
Temperate	`aph(3')-Ia_7_X62115`	0.054	0.071
	`mef(A)_2_U83667`	0.050	0.214
	`lsa(E)_1_JX560992`	0.048	0.305

4.4 Cross-zone Centrality Comparison

The Kruskal-Wallis test on betweenness centrality distributions across the three zones yielded $H = 3.219$, $p = 0.200$ (Figure **??**). This result is not statistically significant at $\alpha = 0.05$, indicating that the *distribution* of betweenness centrality values is statistically similar across zones. The apparent zone differences in hub ARG identity (Section **??**) therefore reflect shifts in *which* ARGs occupy central positions, rather than a change in the overall degree of network centralisation.

Figure 4: Distribution of betweenness centrality across climate zones. Kruskal-Wallis test: $H = 3.22$, $p = 0.20$ (not significant). The three zones share a similar right-skewed centrality distribution; most ARGs have low betweenness while a small number serve as hubs.

5 Discussion

5.1 Climate-stratified ARG network structure

All three co-occurrence networks exhibit properties consistent with other biological co-occurrence networks: sparse but non-trivial density, high clustering coefficients, and modular organisation. The zone-level differences in modularity are particularly informative.

The tropical network's high modularity ($Q = 0.41$) is consistent with the ecological complexity of tropical environments. Tropical climates support high microbial biodiversity, complex soil chemistry, and a wide range of ecological niches spanning rainforest soils, freshwater systems, and coastal sediments. This ecological heterogeneity may produce compartmentalised ARG communities because different microhabitats within tropical zones impose distinct selective pressures, favouring different sets of resistance determinants and mobile genetic elements [**?**]. When samples from ecologically distinct niches are pooled within a single climate zone, the resulting network naturally separates into communities corresponding to those niche-specific resistomes. The high modularity value observed here is thus broadly consistent with tropical zones harbouring a more partitioned, ecologically structured resistome.

The comparatively low modularity of the arid zone ($Q = 0.26$, below the conventional 0.3 threshold) points toward a different pattern of ARG ecology. Arid environments impose a relatively uniform set of stressors: water limitation, high temperatures, and low organic carbon. These conditions may select for a more generalist microbial community with a correspondingly less compartmentalised set of resistance mechanisms [**?**]. An alternative explanation is that arid-zone samples in the mARG dataset are dominated by a small number of environment types, such as desert soil and dryland agricultural sites, which would reduce between-niche variation and compress the network into fewer, less distinct communities. Either way, the low modularity of the arid network suggests that resistance genes in dry climates co-occur in a more integrated, less segregated fashion.

The temperate zone occupies an intermediate position ($Q = 0.31$), reflecting the environmental diversity of the C and D climate classes, which span Mediterranean shrublands, continental grasslands, and boreal forest soils across multiple continents. The temperate zone is also heavily influenced by agricultural antibiotic use, particularly in North America and Europe, where animal husbandry contributes substantially to the resistance gene load in soils and waterways [?]. This anthropogenic input may reinforce certain co-occurrence edges while diluting the natural ecological signal, yielding the intermediate modularity observed.

5.2 Hub ARG identity across zones

The recurrence of `tet(M)` as a top hub in both tropical and arid zones reflects its well-documented ecological breadth. `tet(M)` encodes a ribosomal protection protein that confers resistance to tetracyclines, one of the most widely used antibiotic classes in both human medicine and animal production worldwide. Critically, `tet(M)` is carried on conjugative transposons of the Tn916 family, which are capable of transfer across a wide range of Gram-positive and Gram-negative bacterial genera [?]. This broad-host-range mobility gives `tet(M)` the ecological capacity to co-occur with a diverse set of other resistance genes across phylogenetically distant bacterial hosts, explaining its structural centrality in networks representing ecologically distinct environments. Its recurrence across both tropical and arid zones, despite the substantially different community compositions and modularity structures of those networks, underscores that certain highly mobile ARGs can function as network connectors irrespective of local climate conditions.

In contrast, the temperate zone hub profile is dominated by `lnu(B)` and `aph(6)-Id`, genes conferring resistance to lincosamides and aminoglycosides respectively. Neither gene appeared among the top five hubs in the tropical or arid networks. Lincosamide and aminoglycoside antibiotics are both extensively used in clinical and veterinary settings concentrated in temperate-zone countries, particularly across Europe and North America [?]. The emergence of these genes as network hubs in temperate samples is broadly consistent with selection pressure from heavy antibiotic use reinforcing co-selection and co-occurrence linkages among resistance determinants adapted to high-usage environments. This pattern suggests that the structural centralisation of specific ARGs in co-occurrence networks may in part reflect the regional antibiotic usage profile of the dominant sampling locations within a climate zone.

`lsa(E)`, a gene conferring combined resistance to pleuromutilins, lincosamides, and streptogramins, originally characterised in *Enterococcus* isolates [?], appeared as a top hub in both arid and temperate zones. This cross-zone prevalence is notable because the two zones differ substantially in network density, modularity, and microbial community context. One plausible explanation is that `lsa(E)` resides on broad-host-range plasmids that facilitate its dissemination across phylogenetically and ecologically diverse microbial communities, making it a persistent co-occurrence partner for other resistance genes regardless of climate context. The clinical relevance of `lsa(E)` is increasing, as it has been detected in livestock-associated *Staphylococcus* and *Enterococcus* strains in multiple geographic regions, and its identification as a cross-zone hub in environmental samples warrants attention in AMR surveillance frameworks.

Taken together, the zone-specific hub profiles suggest two coexisting processes in the environmental resistome. First, highly mobile genes with broad host ranges (exemplified by `tet(M)`) generate stable co-occurrence hubs that are structurally central across climatically diverse networks. Second, the identity of additional hubs is shaped by local conditions, including antibiotic use patterns and ecological niche diversity, producing zone-specific centrality signatures. This distinction has practical implications: surveillance strategies targeting structurally central ARGs should prioritise broadly mobile genes as universal targets while also monitoring for the locally dominant hubs that characterise each climate or geographic context.

5.3 Interpretation of non-significant Kruskal-Wallis result

The non-significant Kruskal-Wallis test ($p = 0.20$) should not be interpreted as evidence that climate zone has no effect on ARG network structure. Network-level comparison via overall centrality distribution is a coarse measure; it cannot capture the zone-specific shifts in hub ARG identity documented in Table **??**. Moreover, the heavy overrepresentation of temperate samples (85.7% of the zoned dataset) may have suppressed inter-zone variation by biasing the temperate distribution with very high statistical power relative to tropical and arid. Future work with more balanced zone sampling would provide a more definitive test.

5.4 Limitations

Several limitations bear noting. First, the dataset is heavily skewed toward temperate-zone samples; tropical and arid zones each contribute fewer than 10% of the zoned data, which limits the statistical power of inter-zone comparisons. Second, the 5% presence filter, while necessary for network stability, excluded 94.8% of ARGs, potentially omitting zone-specific rare resistance elements. Third, the Spearman correlation threshold ($r \geq 0.3$) and p-value cutoff (< 0.05) are conventional but arbitrary; network topology is sensitive to these choices. Finally, co-occurrence does not imply physical co-localisation on a mobile genetic element; confounding by shared ecological niche cannot be excluded without confirmatory genomic evidence.

6 Conclusion

This study presents the first large-scale, climate-stratified co-occurrence network analysis of environmental ARGs. Using the mARG dataset and Köppen-Geiger climate zone assignment, we constructed weighted ARG co-occurrence networks for tropical, arid, and temperate zones and characterised their topology through community detection and centrality analysis.

Key findings are: (1) all three zones exhibit non-trivial modular structure; (2) tropical networks are the most compartmentalised ($Q = 0.41$), while arid networks are the most integrated ($Q = 0.26$); (3) `tet(M)` and `lsa(E)` recur as structural hubs across multiple zones, consistent with their known mobility; and (4) overall betweenness centrality distributions are statistically similar across zones despite qualitative differences in hub ARG identity.

These findings contribute to our understanding of how climate context shapes the ecology of antibiotic resistance at the gene-network level and provide a framework for future comparative resistome studies with more geographically balanced sampling.

From a public health and surveillance standpoint, the identification of `tet(M)` and `lsa(E)` as cross-zone structural hubs suggests that monitoring these genes in environmental samples may provide early warning signals of resistome shifts across climatically diverse settings. The zone-specific hub profiles further indicate that universal surveillance frameworks should be complemented by regionally tailored monitoring that accounts for the local antibiotic use patterns and ecological conditions that shape which resistance genes occupy central positions in the co-occurrence network. Future work with geographically balanced sampling and longitudinal datasets will be needed to determine whether the structural differences observed here are stable features of climate-zone resistomes or reflect sampling artefacts in the current mARG dataset.

Data and Code Availability

The mARG dataset used in this study is publicly available at Zenodo (accession 6919377). The Köppen-Geiger climate raster is available at Figshare (dataset 21789074; Beck *et al.* 2023). All analysis scripts are available in the project repository at https://github.com/Jamesardc19/pitch2-arm-gene-network-main.

Author Contributions

J.A.R.D.C. designed and implemented the full analytical pipeline, including data preprocessing, network construction, community detection, centrality analysis, and visualisation. A.C.M.L. contributed to dataset sourcing and literature review for the Background and Methods sections. H.N.M. conducted the literature review and drafted the Introduction and Review of Related Literature. All authors contributed to the Conclusion.

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